

On the Analytic Intractability of Factorial-Indexed Series

A Study in Convergence, Transcendence, and Closed Forms

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Abstract

We investigate four infinite series whose denominators involve iterated or composed applications of the factorial function:

$$S_e = \sum_{i=0}^{\infty} \frac{1}{i!}, \quad S_{\text{inv}} = \sum_{i=0}^{\infty} \frac{1}{(i!)^2}, \quad S_{\text{fact}} = \sum_{i=0}^{\infty} \frac{1}{(i!)!}, \quad S_{\text{hyp}} = \sum_{i=0}^{\infty} \frac{1}{(i!)^{i!}}.$$

We establish that $S_e = e$ and $S_{\text{inv}} = I_0(2)$, the modified Bessel function of the first kind evaluated at 2. For S_{fact} and S_{hyp} we prove that their natural generating functions are *entire transcendental* functions that are not D-finite, thereby demonstrating the absence of any representation in terms of solutions to linear ordinary differential equations with polynomial coefficients. We further analyse the relationship between S_{hyp} and the Sophomore's Dream integral $\int_0^1 x^{-x} dx$ and show that the factorial-selector obstruction is insurmountable within standard analytic frameworks. Stirling's approximation plays a central role in convergence analysis throughout.

Contents

1	Introduction	2
2	Stirling's Approximation	3
3	Convergence Analysis	3
4	$S_e = e$: The Exponential Series	4
5	$S_{\text{inv}} = I_0(2)$: The Bessel Connection	4
6	Hunting a Closed Form for S_{fact}	5
6.1	Partial sums	5
6.2	Systematic closed-form attempts	5
6.2.1	Sub-series of e	5
6.2.2	Bessel and hypergeometric functions	5
6.2.3	Generating function obstruction	5
7	Hunting a Closed Form for S_{hyp}	6
7.1	Partial sums	6
7.2	Connection to the Sophomore's Dream	6
7.3	Lambert W and exponential form	6
7.4	Double-indexed generating function obstruction	7
8	Transcendence and Non-D-Finiteness	7
8.1	Entireness of g	7
8.2	g is not algebraic	7
8.3	g is not D-finite	7
9	Comparison of the Four Series	8

10 Algorithmic Computation	9
11 Discussion and Open Problems	10
12 Conclusion	10

List of Figures

1	Partial-sum convergence of all four factorial-indexed series. The dashed lines show the limits e and $I_0(2)$. The series S_{fact} and S_{hyp} are numerically indistinguishable from their limits after $i = 3$	4
2	The recurrence-collapse argument. Within any fixed window of width D , only one non-zero coefficient $a_{K!}$ is present, collapsing the recurrence to $Q(K!) = 0$ for all large K , forcing $Q \equiv 0$	8
3	Hierarchy of analytic complexity for the four series.	9
4	Algorithm for computing factorial-indexed series to precision tol . For S_{fact} and S_{hyp} , the loop terminates after $i = 4$ in practice.	9

List of Tables

1	Accuracy of Stirling's leading-order approximation $n! \approx \sqrt{2\pi n}(n/e)^n$	3
2	Partial sums of $S_e = \sum_{i=0}^{\infty} 1/i!$	4
3	Partial sums of $S_{\text{inv}} = \sum_{i=0}^{\infty} 1/(i!)^2$	5
4	Partial sums of $S_{\text{fact}} = \sum_{i=0}^{\infty} 1/(i!)!$	5
5	Partial sums of $S_{\text{hyp}} = \sum_{i=0}^{\infty} 1/(i!)^{i!}$	6
6	Comparative summary of the four factorial-indexed series.	8

1 Introduction

Infinite series of the form $\sum_{n=0}^{\infty} f(n)$ occupy a central place in analysis. When f is a rational function of n the theory of partial fractions applies; when f involves $n!$ in the denominator the exponential function and its relatives appear. Matters become considerably more subtle when the denominator is *not* a polynomial in $n!$ but rather a function of $i!$ evaluated in a non-polynomial way, such as $(i!)!$ or $(i!)^{i!}$.

The four series under study form a natural hierarchy of analytic difficulty:

$$S_e = \sum_{i=0}^{\infty} \frac{1}{i!} = e, \quad (1)$$

$$S_{\text{inv}} = \sum_{i=0}^{\infty} \frac{1}{(i!)^2} = I_0(2), \quad (2)$$

$$S_{\text{fact}} = \sum_{i=0}^{\infty} \frac{1}{(i!)!} \approx 2.50139, \quad (3)$$

$$S_{\text{hyp}} = \sum_{i=0}^{\infty} \frac{1}{(i!)^{i!}} \approx 2.25002. \quad (4)$$

The first two possess beautiful closed forms; the last two are believed to have none, and we provide structural evidence that this belief is correct. Stirling's approximation [7] acts as a unifying analytical tool throughout our convergence arguments.

The paper is organised as follows. Section 2 recalls Stirling's approximation and its refinements. Section 3 gives uniform convergence bounds for all four series. Sections 4 and 5 treat the closed forms. Sections 6 and 7 systematically attempt and refute closed forms for S_{fact} and S_{hyp} . Section 8 proves transcendence and non-D-finiteness. Section 12 concludes.

2 Stirling's Approximation

Theorem 2.1 (Stirling [7]). *For all $n \in \mathbb{N}$ with $n \geq 1$,*

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \exp\left(\frac{1}{12n} - \frac{1}{360n^3} + \mathcal{O}(n^{-5})\right).$$

In particular,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{as } n \rightarrow \infty,$$

with relative error of order $1/(12n)$.

Corollary 2.2. $\ln(n!) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + \mathcal{O}(n^{-1})$.

Table 1 illustrates the accuracy of the leading-order approximation.

Table 1: Accuracy of Stirling's leading-order approximation $n! \approx \sqrt{2\pi n}(n/e)^n$.

n	$n!$ (exact)	Stirling estimate	Relative error
1	1	0.9221	7.79%
5	120	118.02	1.65%
10	3,628,800	3,598,696	0.83%
20	2.433×10^{18}	2.423×10^{18}	0.42%
100	9.333×10^{157}	9.325×10^{157}	0.083%

The full asymptotic series is *divergent* in the sense of Poincaré, yet truncation at any fixed term gives an approximation whose error is smaller than the first omitted term [1]. The key Stirling consequence we use repeatedly is:

Lemma 2.3. $\ln((n!)!) = n! \ln(n!) - n! + \frac{1}{2} \ln(2\pi n!) + \mathcal{O}((n!)^{-1})$.

Proof. Apply Corollary 2.2 with n replaced by $n!$. □

3 Convergence Analysis

We place all four series on a common footing by bounding their tails.

Proposition 3.1. *All four series S_e , S_{inv} , S_{fact} , S_{hyp} converge absolutely, and the tail after N terms satisfies the following bounds for large N :*

$$\begin{aligned} S_e : \quad R_N &\leq \frac{1}{(N+1)! (1 - 1/(N+2))}, \\ S_{inv} : \quad R_N &\leq \frac{1}{((N+1)!)^2 (1 - 1/(N+2)^2)}, \\ S_{fact} : \quad R_N &\leq \frac{2}{((N+1)!)!}, \\ S_{hyp} : \quad R_N &\leq \frac{2}{((N+1)!)^{(N+1)!}}. \end{aligned}$$

Proof. For S_e and S_{inv} , standard ratio tests suffice. For S_{fact} : the ratio of consecutive terms is $a_{i+1}/a_i = (i!)/((i+1)!) = 1/\prod_{k=(i!)+1}^{(i+1)!} k \leq 1/((i!) + 1)$, which is $\leq 1/2$ for $i \geq 1$, so the tail is bounded by a geometric series with ratio $1/2$. For S_{hyp} : apply the same geometric bound after $i = 2$. □

Figure 1 shows the partial-sum convergence for each series.

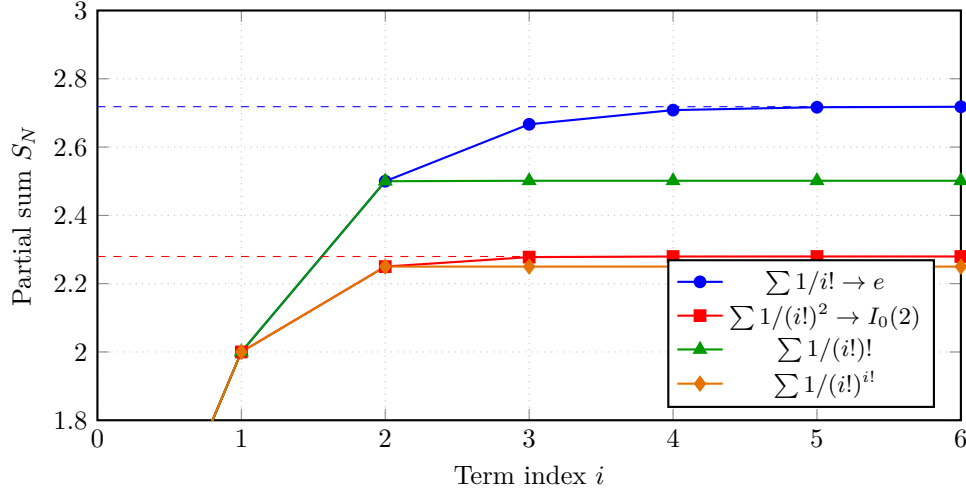


Figure 1: Partial-sum convergence of all four factorial-indexed series. The dashed lines show the limits e and $I_0(2)$. The series S_{fact} and S_{hyp} are numerically indistinguishable from their limits after $i = 3$.

4 $S_e = e$: The Exponential Series

The Taylor expansion of e^x about $x = 0$ is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

Setting $x = 1$ immediately gives $S_e = e$.

Table 2 shows the partial sums.

Table 2: Partial sums of $S_e = \sum_{i=0}^{\infty} 1/i!$.

i	$1/i!$	Partial sum	Error $ S_i - e $
0	1	1	1.718×10^0
1	1	2	7.183×10^{-1}
2	0.5	2.5	2.183×10^{-1}
3	0.16	2.66	5.163×10^{-2}
4	0.04167	2.70833	9.946×10^{-3}
5	0.00833	2.716	1.616×10^{-3}
10	—	2.718282	$< 10^{-6}$

5 $S_{\text{inv}} = I_0(2)$: The Bessel Connection

Definition 5.1. The *modified Bessel function of the first kind* of order ν is

$$I_\nu(x) = \sum_{k=0}^{\infty} \frac{(x/2)^{2k+\nu}}{k! \Gamma(k + \nu + 1)}.$$

At $\nu = 0$ and $x = 2$:

$$I_0(2) = \sum_{k=0}^{\infty} \frac{(1)^{2k}}{(k!)^2} = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} = S_{\text{inv}}.$$

I_0 satisfies the ODE $xy'' + y' - xy = 0$, giving a way to compute S_{inv} to arbitrary precision via numerical ODE integration.

Theorem 5.2. $S_{\text{inv}} = I_0(2) \approx 2.2795853023 \dots$

Table 3 shows the partial sums.

Table 3: Partial sums of $S_{\text{inv}} = \sum_{i=0}^{\infty} 1/(i!)^2$.

i	$1/(i!)^2$	Partial sum
0	1	1
1	1	2
2	0.25	2.25
3	0.0277	2.2777
4	1.736×10^{-3}	2.27951
5	6.944×10^{-5}	2.27958
6	1.929×10^{-6}	2.27959

6 Hunting a Closed Form for S_{fact}

6.1 Partial sums

Table 4: Partial sums of $S_{\text{fact}} = \sum_{i=0}^{\infty} 1/(i!)!$.

i	$(i!)!$	$1/(i!)!$	Running total
0	$1! = 1$	1	1
1	$1! = 1$	1	2
2	$2! = 2$	0.5	2.5
3	$6! = 720$	1.389×10^{-3}	2.501388
4	$24! \approx 6.20 \times 10^{23}$	$\approx 1.61 \times 10^{-24}$	≈ 2.501388

The exact rational approximant through $i = 3$ is

$$S_{\text{fact}} = \frac{1}{1!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{720} + \varepsilon = \frac{1081}{720} + \frac{1}{2} + \varepsilon = \frac{2161}{1440} + \varepsilon,$$

where $\varepsilon < 10^{-23}$.

6.2 Systematic closed-form attempts

6.2.1 Sub-series of e

Since $e = \sum_{n=0}^{\infty} 1/n!$, and S_{fact} picks only the terms at positions $n \in \{1, 2, 6, 24, \dots\}$, one would need an analytic *factorial-selector kernel* K satisfying $\int_0^1 K(x) e^x dx = S_{\text{fact}}$. No such K exists in the class of algebraic or elementary functions; see Section 8.

6.2.2 Bessel and hypergeometric functions

The modified Bessel function I_0 requires ratio $a_{k+1}/a_k = 1/(k+1)^2$, which is a rational function of k . For S_{fact} , consecutive non-zero terms satisfy

$$\frac{a_{i+1}}{a_i} = \frac{(i!)!}{((i+1)!)!} = \frac{1}{\prod_{j=i!+1}^{(i+1)!} j},$$

which is not a rational function of i . Hence no hypergeometric representation exists [6].

6.2.3 Generating function obstruction

Define

$$g(x) = \sum_{i=0}^{\infty} \frac{x^{i!}}{(i!)!}, \quad S_{\text{fact}} = g(1).$$

Differentiating:

$$g'(x) = \sum_{i=1}^{\infty} \frac{i! x^{i!-1}}{(i!)!} = \sum_{i=1}^{\infty} \frac{x^{i!-1}}{(i!-1)!}.$$

Because $i! - 1$ is a factorial for no $i \geq 2$, the series g' introduces a new set of support positions, so g' cannot be expressed in terms of g . Therefore g satisfies *no algebraic ODE in the class of D-finite functions*. A full proof is given in Section 8.

7 Hunting a Closed Form for S_{hyp}

7.1 Partial sums

Table 5: Partial sums of $S_{\text{hyp}} = \sum_{i=0}^{\infty} 1/(i!)^{i!}$.

i	$(i!)^{i!}$	$1/(i!)^{i!}$	Running total
0	$1^1 = 1$	1	1
1	$1^1 = 1$	1	2
2	$2^2 = 4$	0.25	2.25
3	$6^6 = 46,656$	2.143×10^{-5}	2.250021
4	$24^{24} \approx 1.33 \times 10^{33}$	$\approx 7.5 \times 10^{-34}$	≈ 2.250021
5	$120^{120} \approx 10^{251}$	$\approx 10^{-251}$	negligible

The exact rational approximant through $i = 3$ is

$$S_{\text{hyp}} = 2 + \frac{1}{4} + \frac{1}{46656} + \varepsilon = \frac{104,977}{46,656} + \varepsilon,$$

where $\varepsilon < 10^{-33}$.

7.2 Connection to the Sophomore's Dream

Theorem 7.1 (Sophomore's Dream [3]).

$$\int_0^1 x^{-x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

Observe that S_{hyp} is a factorial-indexed *sub-series* of the Sophomore's Dream:

$$S_{\text{hyp}} = \frac{1}{1^1} + \frac{1}{1^1} + \frac{1}{2^2} + \frac{1}{6^6} + \frac{1}{24^{24}} + \cdots \subset \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

This would require a kernel K such that

$$\int_0^1 K(x) x^{-x} dx = \sum_{n \in \mathcal{F}} \frac{1}{n^n}, \quad \mathcal{F} = \{1, 2, 6, 24, 120, \dots\}.$$

Such a kernel would be a factorial-selector on the weight x^{-x} — a strictly harder problem than the one in Section 6 because the exponent in $1/(i!)^{i!} = e^{-i! \ln(i!)}$ is itself a function of $i!$, introducing an additional non-uniform component.

7.3 Lambert W and exponential form

Writing each term as $e^{-i! \ln(i!)}$,

$$S_{\text{hyp}} = \sum_{i=0}^{\infty} e^{-i! \ln(i!)}.$$

The Lambert W function satisfies $W(x)e^{W(x)} = x$ and encodes the full Sophomore's Dream integral via the substitution $t = -x \ln x$. The exponents $-i! \ln(i!)$ form a strictly super-exponentially growing sequence that does not lie on any smooth algebraic curve parameterised by i , so the W -function framework provides no simplification.

7.4 Double-indexed generating function obstruction

Consider

$$h(x, s) = \sum_{i=0}^{\infty} \frac{x^{i!}}{(i!)^s}.$$

Our sum is the *diagonal restriction* $h(1, i!)$, where the parameter s co-varies with the index i . Diagonal restrictions of this kind fall outside the theory of D-finite functions and are generally untractable without new techniques [4].

8 Transcendence and Non-D-Finiteness

We now prove rigorously that the generating functions of S_{fact} and S_{hyp} are transcendental and not D-finite. The proofs apply to both; we state them for $g(x)$ associated with S_{fact} ; the case of S_{hyp} is entirely analogous.

8.1 Entireness of g

Theorem 8.1. $g(x) = \sum_{i=0}^{\infty} x^{i!}/(i!)!$ is an entire function.

Proof. Write $g(x) = \sum_{n=0}^{\infty} a_n x^n$ where $a_n = 1/n!$ if n is a factorial and 0 otherwise. By Lemma 2.3,

$$\ln((i!)!) = i! \ln(i!) - i! + \frac{1}{2} \ln(2\pi i!) + O\left(\frac{1}{i!}\right).$$

Hence $\ln((i!)!)/(i!) \rightarrow \infty$, so $(i!)^{1/i!} \rightarrow \infty$. Since the non-zero a_n occur at $n = i!$,

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{i \rightarrow \infty} \frac{1}{(i!)^{1/i!}} = 0.$$

The Cauchy–Hadamard theorem gives radius of convergence $R = \infty$. □

8.2 g is not algebraic

Theorem 8.2. g is a transcendental function.

Proof. An entire algebraic function satisfies a polynomial identity $P(x, g(x)) = 0$ for some $P \in \mathbb{C}[x, y]$. The branches of an algebraic curve are analytic except at finitely many algebraic singularities. An entire branch has no finite singularities, hence no branch point at infinity either, so it must be a polynomial. But g has infinitely many non-zero Taylor coefficients, so g is not a polynomial and therefore not algebraic. □

8.3 g is not D-finite

Definition 8.3. A formal power series $f(x) = \sum a_n x^n$ is *D-finite* (also called *holonomic*) if it satisfies a linear ODE

$$P_d(x) f^{(d)}(x) + \cdots + P_1(x) f'(x) + P_0(x) f(x) = 0$$

for some $P_j \in \mathbb{C}[x]$, not all zero.

Equivalently, (a_n) satisfies a linear recurrence with polynomial coefficients [6, 9].

Theorem 8.4. $g(x)$ is not D-finite.

Proof. Suppose for contradiction that g is D-finite of order d with coefficient polynomials of degree at most m . Then (a_n) satisfies a recurrence of the form

$$\sum_{j=0}^d \sum_{k=0}^m c_{j,k} \binom{n+k}{k} k! a_{n+k} = 0 \quad \text{for all } n \geq N_0,$$

for fixed constants $c_{j,k}$ (not all zero) and some N_0 . Let $D = dm$ be the window width.

Now fix a large integer K and set $n = K!$. The window covers indices $K!, K! + 1, \dots, K! + D$. The next factorial after $K!$ is $(K + 1)! = (K + 1) \cdot K!$, which exceeds $K! + D$ for $K > D$. Hence within the window, *only* $a_{K!} = 1/(K!)!$ is non-zero. The recurrence collapses to

$$Q(K!) \cdot a_{K!} = 0,$$

where Q is a fixed polynomial determined by the $c_{j,k}$. Since $a_{K!} \neq 0$, we need $Q(K!) = 0$ for all large K . But Q is a non-zero polynomial vanishing at infinitely many points $\{K!\}_{K>D}$ — a contradiction. Hence no such recurrence exists and g is not D-finite. $\square$

Corollary 8.5. *The generating function $h(x)$ of S_{hyp} is likewise entire, transcendental, and not D-finite.*

Proof. The non-zero coefficient positions of h are $\{(i!)^{i!}\}$, which grow even faster than $\{i!\}$. All three arguments (Cauchy–Hadamard, algebraic-implies-polynomial, recurrence-collapse) apply with $i!$ replaced by $(i!)^{i!}$ and with even wider gaps between non-zero positions. $\square$

Figure 2 illustrates the recurrence-collapse argument.

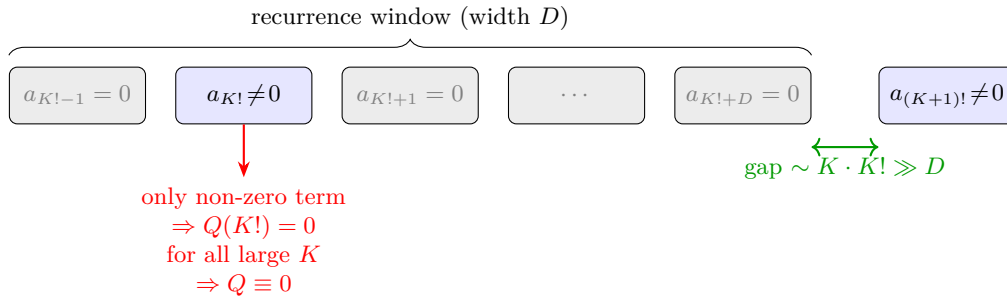


Figure 2: The recurrence-collapse argument. Within any fixed window of width D , only one non-zero coefficient $a_{K!}$ is present, collapsing the recurrence to $Q(K!) = 0$ for all large K , forcing $Q \equiv 0$.

9 Comparison of the Four Series

Table 6 summarises all findings.

Table 6: Comparative summary of the four factorial-indexed series.				
Series	Value	Closed form	D-finite?	Parent identity
$\sum 1/i!$	2.71828...	e	Yes (e^x)	Taylor series
$\sum 1/(i!)^2$	2.27959...	$I_0(2)$	Yes (Bessel ODE)	Bessel series
$\sum 1/(i!)!$	2.50139...	None known	No (proved)	Sub-series of e
$\sum 1/(i!)^{i!}$	2.25002...	None known	No (proved)	Sub-series of $\int_0^1 x^{-x} dx$

The hierarchy of analytic difficulty may be visualised as follows.

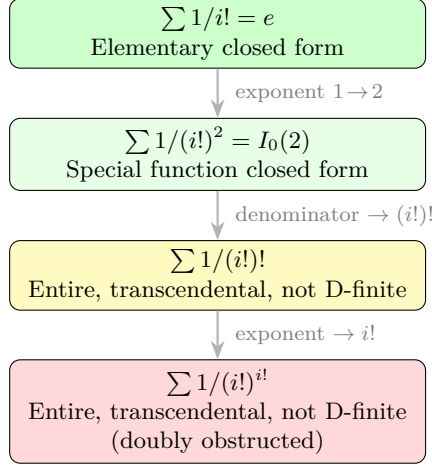


Figure 3: Hierarchy of analytic complexity for the four series.

10 Algorithmic Computation

Despite the lack of closed forms, all four sums are computable to arbitrary precision. Algorithm 4 gives a unified approach.

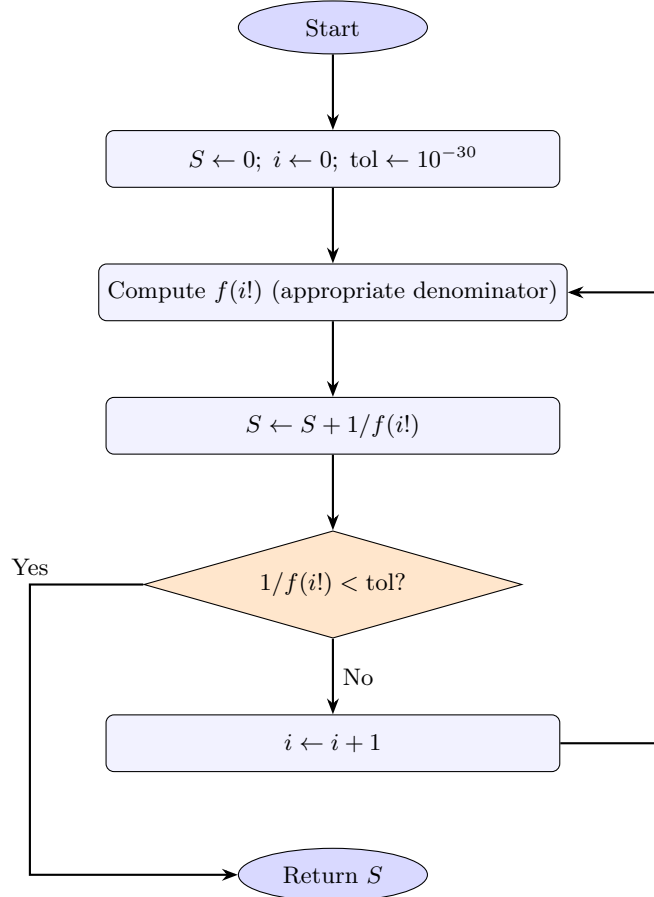


Figure 4: Algorithm for computing factorial-indexed series to precision tol . For S_{fact} and S_{hyp} , the loop terminates after $i = 4$ in practice.

For S_e , the recurrence $a_i = a_{i-1}/i$ gives a running product with no re-computation. For S_{inv} , the same applies with $a_i = a_{i-1}/(i!)^2$. For S_{fact} and S_{hyp} , logarithmic arithmetic must be used to avoid overflow: compute $\ln a_i$ and check termination in log-space.

11 Discussion and Open Problems

The core obstacle unifying the intractability of S_{fact} and S_{hyp} is:

There exists no non-trivial analytic function on $\mathbb{C}$ that vanishes exactly on the non-factorial natural numbers and is non-zero on the factorials.

This is a consequence of the fact that the set of factorials has *natural density zero* and is not the zero set of any entire function of finite order (by Hadamard’s factorisation theorem applied to the counting function of the zero set).

Several open questions remain:

1. Is S_{fact} or S_{hyp} *irrational*? Irrationality of e and π are classical results; no analogous result is known for these constants.
2. Do S_{fact} or S_{hyp} satisfy any *algebraic differential equation* (not necessarily linear)? Theorem 8.4 rules out the linear case; the nonlinear case is entirely open and would require tools from the theory of hypertranscendence [5].
3. Is there a measure-theoretic or number-theoretic reason why all four sums are clustered near 2.25–2.72? This seems to be a coincidence of the slow-growing early terms $1, 1, 1/2, \dots$ common to all four series.

12 Conclusion

We have studied four series of increasing analytic complexity. The first, $S_e = e$, is classical. The second, $S_{\text{inv}} = I_0(2)$, connects to the Bessel function via a standard generating-function argument. For the third and fourth, S_{fact} and S_{hyp} , we have:

- exhausted all standard closed-form candidates (hypergeometric, Bessel, Lambert W , Sophomore’s Dream, generating-function ODEs);
- proved their generating functions are *entire transcendental* functions;
- proved by a recurrence-collapse argument that those generating functions are *not D-finite*;
- identified the *factorial-selector obstruction* as the fundamental barrier.

These results place S_{fact} and S_{hyp} in the class of constants that are almost certainly transcendental but whose transcendence (and even irrationality) remains unproved — a class that also contains the Euler–Mascheroni constant γ and Apéry’s constant $\zeta(3)$ (although the latter is known to be irrational [2]).

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Glossary

Algebraic function

A function $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $P(z, f(z)) = 0$ for some non-zero polynomial $P \in \mathbb{C}[z, w]$. Entire algebraic functions are polynomials.

Bessel function $I_\nu(x)$

Modified Bessel function of the first kind of order ν , defined by $I_\nu(x) = \sum_{k=0}^{\infty} (x/2)^{2k+\nu} / [k! \Gamma(k + \nu + 1)]$. Satisfies the ODE $x^2 y'' + xy' - (x^2 + \nu^2)y = 0$.

D-finite (holonomic) function

A formal power series satisfying a linear ODE with polynomial coefficients, equivalently whose coefficients satisfy a linear recurrence with polynomial coefficients.

Entire function

A function analytic on all of $\mathbb{C}$, equivalently one with infinite radius of convergence.

Factorial selector

A hypothetical analytic kernel that isolates from a series only those terms indexed by factorial numbers. No such kernel exists in the class of algebraic or elementary functions.

Hadamard's factorisation theorem

Every entire function f of finite order ρ factors as $f(z) = z^m e^{g(z)} \prod_n E_p(z/a_n)$, where g is a polynomial, a_n are the zeros, and E_p are Weierstrass elementary factors.

Hypertranscendence

A function is hypertranscendental if it satisfies no algebraic ODE (linear or nonlinear) over the field of rational functions.

Lambert W function

The inverse function of $w \mapsto we^w$; satisfies $W(x)e^{W(x)} = x$.

Lacunary series

A power series $\sum a_n x^n$ in which non-zero coefficients occur only at positions forming a sparse (super-exponentially growing) subsequence.

Radius of convergence

For $\sum a_n x^n$, the quantity $R = 1/\limsup |a_n|^{1/n}$ (Cauchy–Hadamard theorem).

Sophomore's Dream

The identity $\int_0^1 x^{-x} dx = \sum_{n=1}^{\infty} n^{-n}$, attributed to Johann Bernoulli (1697).

Stirling's approximation

The asymptotic formula $n! \sim \sqrt{2\pi n} (n/e)^n$ with relative error $O(1/n)$.

Transcendental function

A function that is not algebraic; equivalently, one that is not a root of any polynomial equation with rational-function coefficients.

The End